Class 5: Data Visualization

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Plotting in R

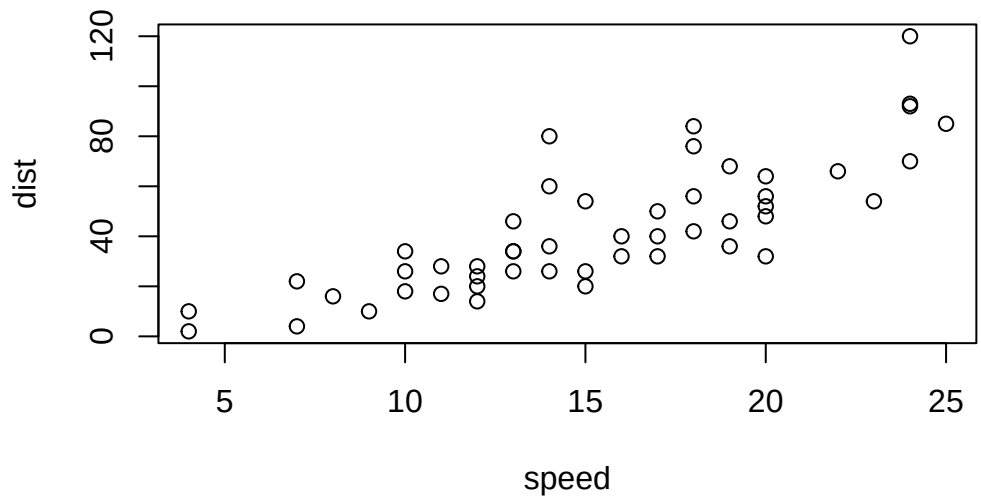
R has many plotting and visualization systems including "base" R.

```
head(cars, n=6)
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

A base plot

```
plot(cars)
```



Base R plots can be quite simple for basic plots when compared to systems like ggplot.

To use an add on package, like ggplot, I have to first get it on my computer - i.e. install it!

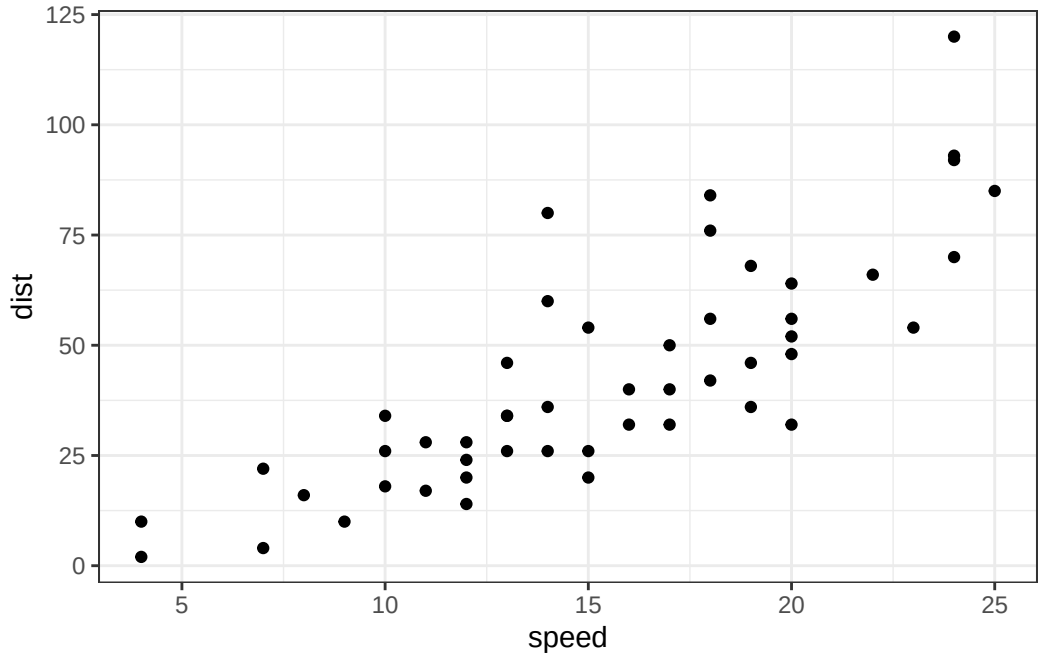
We use the function `install.packages()` with the name of the package we want to install.

```
library(ggplot2)
ggplot(cars)
```

ggplot is much more verbose than base R plot and every single ggplot needs at least 3 things:

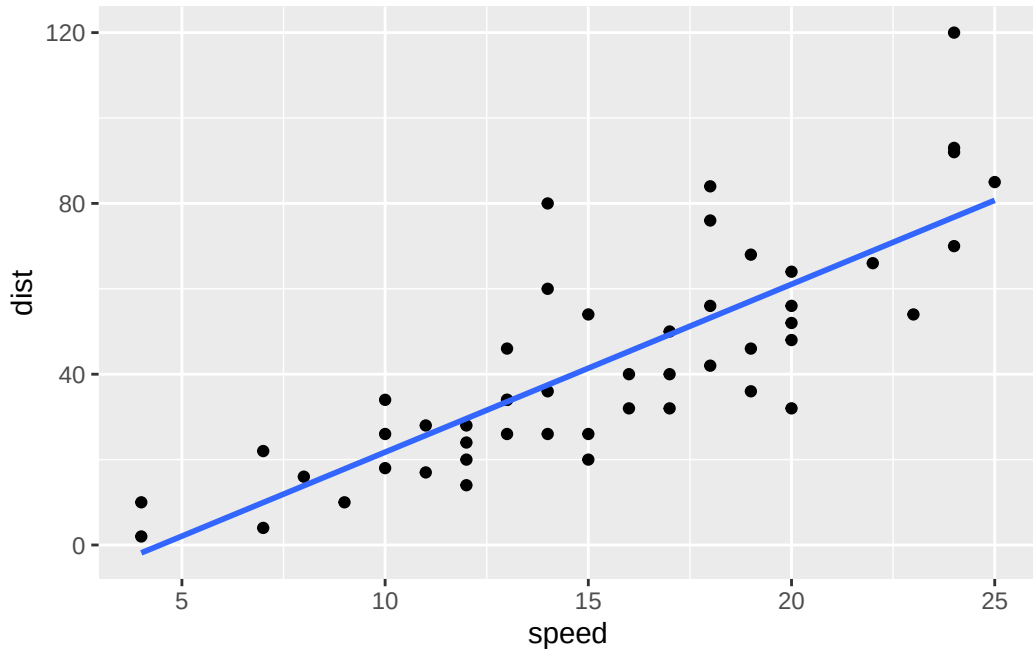
- **Data** (this is the data.frame with the stuff we want to plot)
- **Aesthetics** or aes() for short (how the data map to the plot)
- **Geoms** (like geom_point(), geom_line() the plot type)

```
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point() +
  theme_bw()
```




```
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point() +
  geom_smooth(se=FALSE, method="lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```



A plot of some gene expresion data

The code to read the data:

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"

genes <- read.delim(url)

head(genes)
```

```
        Gene Condition1 Condition2      State
1      A4GNT -3.6808610 -3.4401355 unchanging
2       AAAS  4.5479580  4.3864126 unchanging
3      AASDH  3.7190695  3.4787276 unchanging
4       AATF  5.0784720  5.0151916 unchanging
5       AATK  0.4711421  0.5598642 unchanging
6 AB015752.4 -3.6808610 -3.5921390 unchanging
```

Q. How many genes are in this dataset?

```
nrow(genes)
```

```
[1] 5196
```

There are 5196 genes in this dataset.

How many genes are up-regulated?

```
table( genes$State )
```

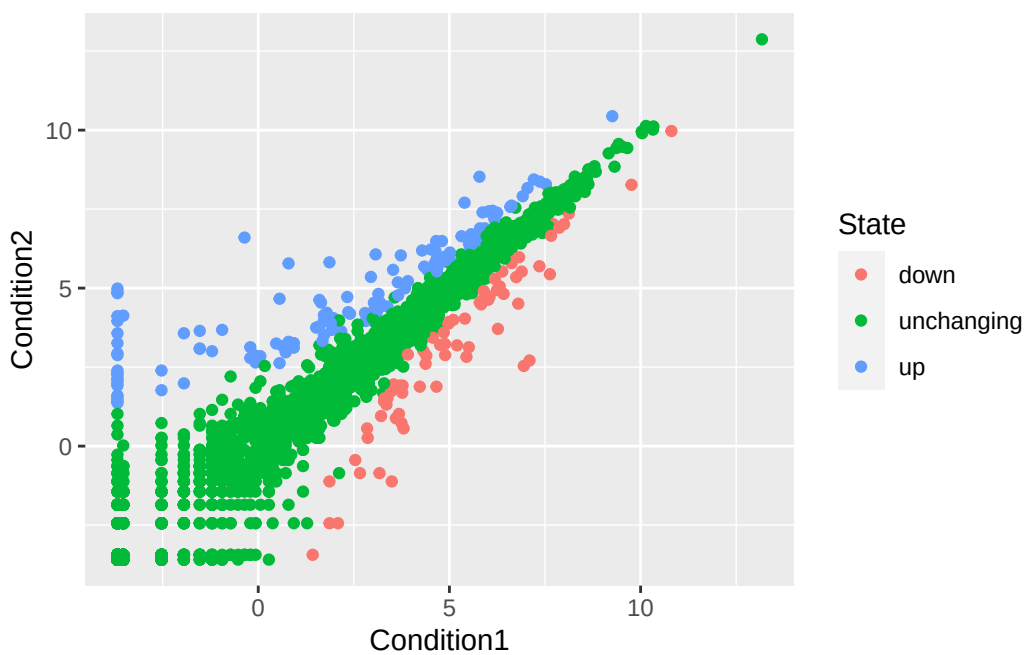
```
      down unchanging          up 
        72       4997         127 
```

```
sum(genes$State == "up")
```

```
[1] 127
```

Let's stop messing and plot it

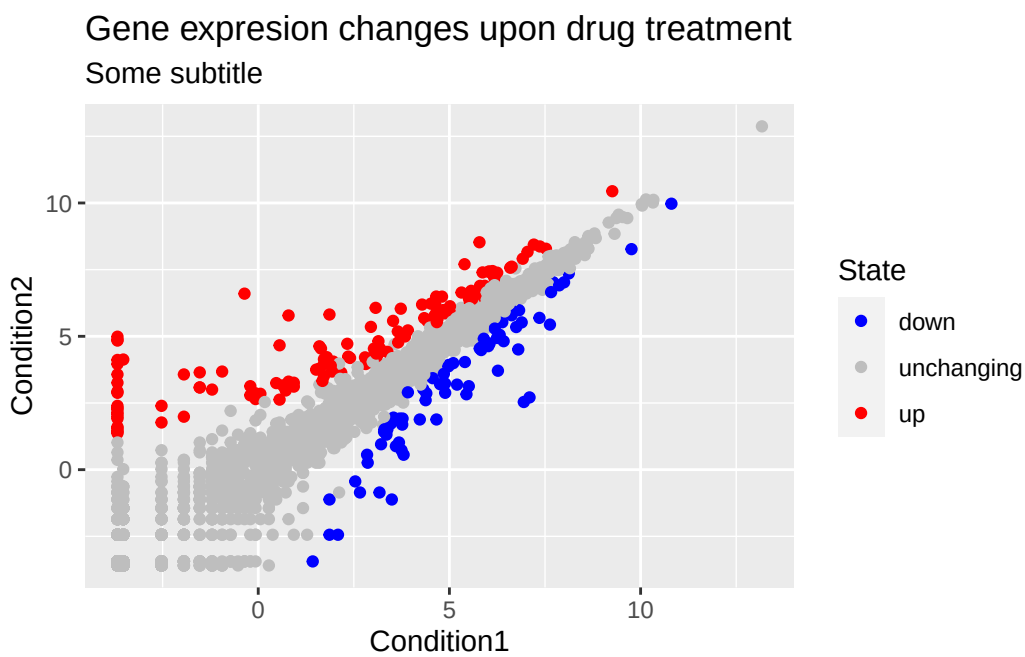
```
ggplot(genes) +
  aes(x=Condition1, y=Condition2, color=State) +
  geom_point()
```



I can save any ggplot object for use later so I don't need to type it all out again. Here I save my starting plot to the object `p` then I can add layers to `p` later on.

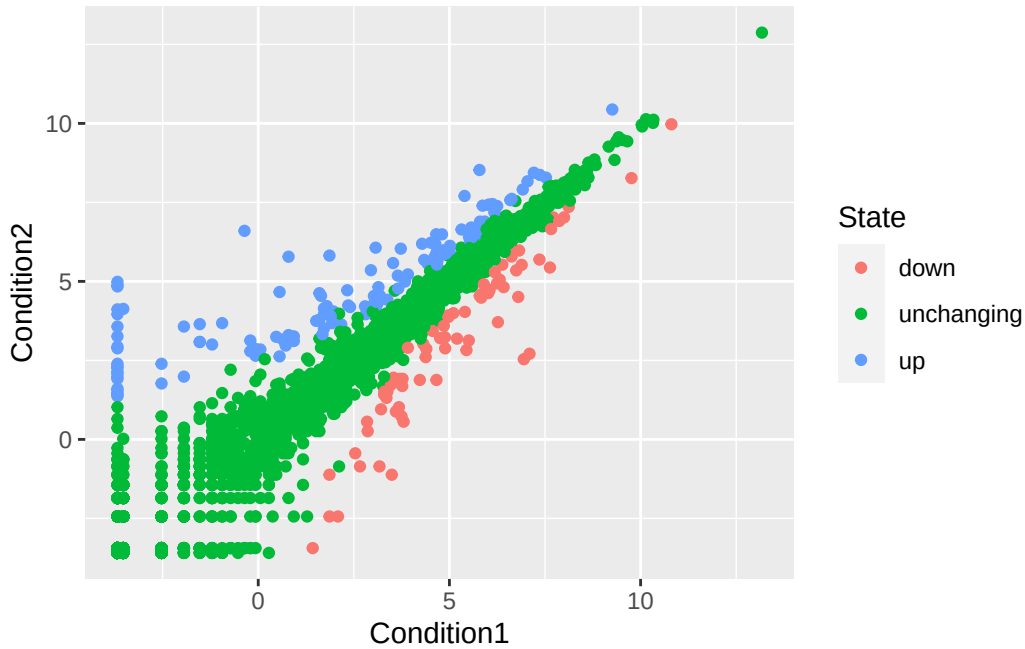
```
p <- ggplot(genes) +
    aes(x=Condition1, y=Condition2, col=State) +
    geom_point()
```

```
p + scale_colour_manual( values=c("blue","gray","red") ) +
  labs(title="Gene expresion changes upon drug treatment",
       subtitle = "Some subtitle")
```

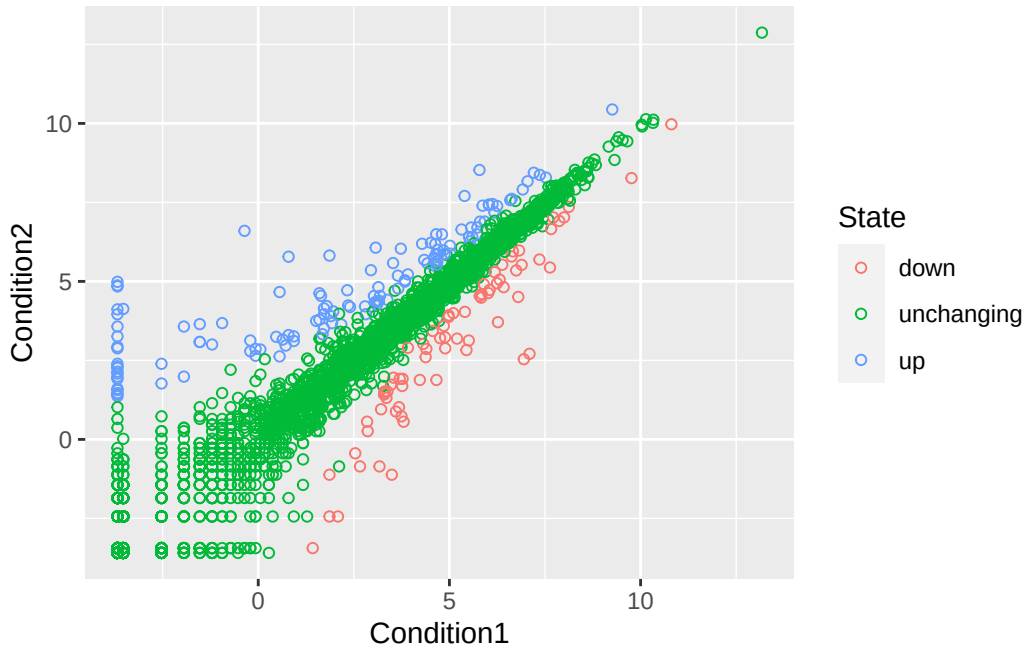


My object `p` is just the same.

```
p
```




```
ggplot(genes) +
  aes(x=Condition1, y=Condition2, color=State) +
  geom_point(shape=21)
```



A more complex ggplot example

One of the big wins with ggplot is how easy it is to facet your data into sub-plots..

Read the gapminder dataset from online

```
# File location online
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.

gapminder <- read.delim(url)
```

and have a wee peak

```
head(gapminder)
```

```
      country continent year lifeExp      pop gdpPercap
1 Afghanistan      Asia 1952  28.801  8425333  779.4453
2 Afghanistan      Asia 1957  30.332  9240934  820.8530
3 Afghanistan      Asia 1962  31.997 10267083  853.1007
4 Afghanistan      Asia 1967  34.020 11537966  836.1971
5 Afghanistan      Asia 1972  36.088 13079460  739.9811
6 Afghanistan      Asia 1977  38.438 14880372  786.1134
```

Q. How many countrys are in this dataset?

```
length( unique(gapminder$country) )
```

```
[1] 142
```

Q. How many years do we have data for?

```
min(gapminder$year)
```

```
[1] 1952
```

```
max(gapminder$year)
```

```
[1] 2007
```

```
range(gapminder$year)
```

```
[1] 1952 2007
```

Q. Which country has the smallest population in the dataset?

```
min(gapminder$pop)
```

```
[1] 60011
```

First where is this min value in the pop vector

```
ind <- which.min(gapminder$pop)
```

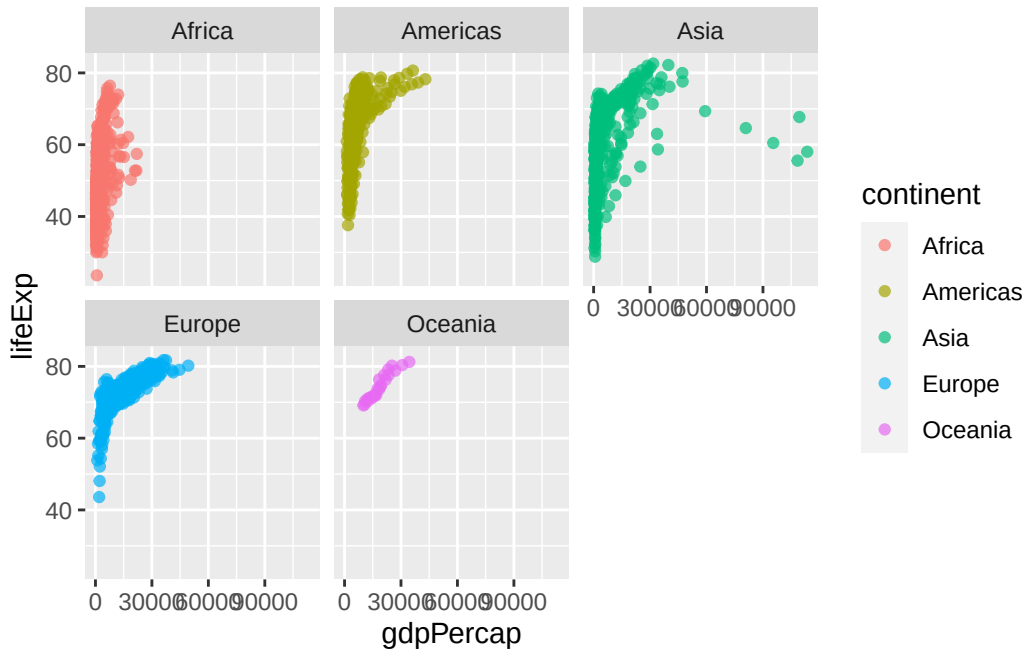
Now use this to access the $country value for this position

```
gapminder$country[ind]
```

```
[1] "Sao Tome and Principe"
```

```r
gapminder[ind,]
```

```
                   country continent year lifeExp   pop gdpPercap
1297 Sao Tome and Principe    Africa 1952  46.471 60011  879.5836
```

Make a first plot of gdb vs life exp

```r
ggplot(gapminder) +
  aes(x=gdpPercap, y=lifeExp, color=continent) +
  geom_point(alpha=0.7) +
  facet_wrap(~continent)
```